25900243

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTS201 - GENERAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. Antonie van Leeuwenhoek, the first person to observe microorganisms under a microscope, described them as **Animalcules**. [K] / [CO1]
2. Media that contain complex ingredients such as peptone, beef extract, or yeast extract are called **complex (undefined)** media. [U] / [CO2, CO3]
3. Viruses that infect bacteria are known as **bacteriophages**. [U] / [CO1]
4. Facultative anaerobes can grow in the presence or absence of **oxygen**. [U] / [CO3]
5. The infectious proteinaceous agents are called **prions**. [U] / [CO1]
6. The primary resource for the classification and identification of bacteria, often referred to as the 'bible of bacteriology', is **Bergey's manual of Systematic bacteriology**. [K] / [CO1, CO2]
7. The enzyme responsible for transposition is **transposase .** [U] / [CO5]
8. Bacteria capable of natural transformation are said to be **competent** [U] / [CO5]
9. The enzyme **lactate dehydrogenase** converts pyruvate to lactate in lactic acid fermentation. [U] / [CO3, CO5]

10. In the disc diffusion method, the zone of inhibition indicates the **Susceptibility** of bacteria to the antibiotic. [U] / [CO3, CO4]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Differentiate between bacterial chromosome and plasmids. [U] / [CO5]

Plasmid DNA is naked without the presence of histone proteins. It is coated with histone proteins. Plasmids are separated from bacterial genomic DNA. Chromosomal DNA freely floats in the cytoplasm of bacterial cells while in eukaryotic organisms, they are found inside the nucleus.

12. Name the four typical phases of a bacterial growth curve. [U] / [CO2, CO3]

The four stages of the microbial growth curve—lag, exponential, stationary, and death phase.

13. Compare natural, semi-synthetic, and synthetic antibiotics. [U] / [CO3, CO4]

Natural antibiotics are produced directly by microorganisms such as bacteria and fungi; examples include penicillin from *Penicillium* and streptomycin from *Streptomyces*.
Semi-synthetic antibiotics are derived from natural antibiotics that are chemically modified in the laboratory to improve their effectiveness, stability, or spectrum of activity; examples include ampicillin and amoxicillin.
Synthetic antibiotics are entirely man-made through chemical synthesis, such as sulfonamides and ciprofloxacin. While natural antibiotics are purely microbial products, semi-synthetic ones combine natural and chemical methods, and synthetic antibiotics are fully artificial, allowing greater control over their properties and action.

14. What is the principle of phase contrast microscopy? [K] / [CO1, CO2]

When light passes through cells, small phase shifts occur, which are invisible to the human eye. In a phase-contrast microscope, these phase shifts are converted into changes in amplitude, which can be observed as differences in image contrast.

15. Describe the mechanism of action of sulfonamides. [U] / [CO3, CO4]

Sulfonamides are synthetic bacteriostatic antibiotics that inhibit bacterial growth by interfering with the synthesis of folic acid**,** an essential vitamin required for DNA, RNA, and protein synthesis in bacteria.

16. Differentiate between F^+ and F^- cells. [U] / [CO5]

The F plasmid, or fertility plasmid, plays a crucial role in bacterial conjugation by enabling the transfer of genetic material between bacteria. F+ cells, which contain the entire F plasmid, act as donor cells. They form an F pilus, a structure that connects to F- recipient cells, which lack the full F plasmid.

17. What is a chemostat? [U] / [CO2, CO3]

A chemostat is a type of continuous culture system used to grow microorganisms under steady-state conditions. A chemostat is a bioreactor in which fresh nutrient medium is continuously added at a constant rate, while an equal volume of culture (containing cells and waste products) is simultaneously removed.

18. State two factors that influence the effectiveness of a disinfectant. [U] / [CO3]

Two factors that influence the effectiveness of a disinfectant are:

1. Concentration of the disinfectant – Higher concentrations generally increase effectiveness up to an optimal level.
2. Contact time – The longer the disinfectant remains in contact with microorganisms, the greater its ability to kill or inhibit them.

19. Name the three sugars present in TSI agar. [K] / [CO1, CO2]

TSI Agar contains three sugars (dextrose, lactose and sucrose), phenol red for detecting carbohydrate fermentation and ferrous ammonium sulfate for detection of hydrogen sulfide production (indicated by blackening in the butt of the tube).

20. How R plasmids confer resistance to bacteria. [U] / [CO5]

R plasmids make bacteria resistant by carrying genes that help destroy the antibiotic block its entry, pump it out, or change the target site — allowing bacteria to survive even in the presence of antibiotics.

21. What is the chemical composition of peptidoglycan? [U] / [CO1]

A peptidoglycan monomer consists of two joined amino sugars, N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM), with a pentapeptide coming off of the NAM. In S. aureus, the pentapeptide consists of the amino acids L-alanine, D-glutamine, L-lysine, and two D-alanines.

22. Mention two nutritional factors essential for microbial growth. [K] / [CO3]

Microorganisms (or microbes) vary significantly in terms of the source, chemical form, and amount of essential elements they need. Some examples of these essential nutrients are carbon, oxygen, hydrogen, phosphorus, and sulfur

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Define sterilization. Discuss in detail the various physical agents of sterilization with examples. [U] / [CO3]

Sterilization is the complete destruction or removal of all forms of microorganisms, including spores, from an object or medium, making it completely free from life.
Physical methods of sterilization- Heat, Filtration, Radiation.

24. Explain the steps involved in specialized transduction. [U] / [CO5]

Specialized transduction is a type of bacteriophage-mediated gene transfer in which a specific portion of the bacterial DNA is transferred from one bacterium to another through a temperate phage (such as lambda phage in *E. coli*).
Phage infects → integrates into host DNA (prophage)
Faulty excision → picks up nearby bacterial gene
New phage carries bacterial DNA → infects another cell
Gene recombines → genetic change in recipient

25. Describe chemical agents of sterilization with examples. [U] / [CO3]

Chemical agents of sterilization are chemical substances used to destroy or eliminate all forms of microorganisms, including spores, on inanimate objects and surfaces. They are used when heat or physical methods are not suitable.

Common examples include alcohols, aldehydes, ethylene oxide, halogens, phenols, and peroxides**,**

26. Write down the principle and procedure of Gram staining technique. [U] / [CO1, CO2]

The Gram staining technique, developed by Hans Christian Gram in 1884**,** is used to differentiate bacteria into two groups **—** Gram-positive and Gram-negative **—** based on differences in their cell wall composition**.**

27. Describe the various techniques for isolation of pure cultures. [U] / [CO2, CO3]

A pure culture may be isolated by the use of special media with specific chemical or physical agents that allow the enrichment or selection of one organism over another. Pure culture isolation techniques such as streak plate, pour plate, spread plate, serial dilution, and enrichment culture help separate individual microbial species from mixed populations for study and identification.

28. Write notes on transport media with examples. [U] / [CO3]

Transport media are those culture media that are used to maintain the viability of pathogenic microorganisms present in a clinical sample and to prevent the potential pathogens to be overgrown by other commensals or contaminants in the specimen during the transportation of the clinical specimen to the clinical/diagnostic laboratory.

Examples: Stuart's medium, Amies medium, Cary–Blair medium, and Viral Transport Medium.

29. Write a short note on the industrial applications of yeast. [K] / [CO1]

Yeast has several crucial industrial applications. Important examples include: Baking Industry: Used as a leavening agent to make bread, cakes, and other bakery products rise. Brewing Industry: Essential for the fermentation process that produces alcoholic beverages like beer and wine.

[4x5 = 20]